

Virtual Reality Experience ‘The World of Hiroshige’

An Immersive Virtual Reality Experience of the World that Inspired the Works of the Japanese Artist

Utagawa Hiroshige

Alex Weight

UTS Animal Logic Academy,
University of Technology, Australia
alex.weight@uts.edu.au

Dylan Neil

UTS Animal Logic Academy,
University of Technology, Australia
dylan.neill@uts.edu.au

Daniel Flood

UTS Animal Logic Academy,
University of Technology, Australia
Daniel.flood@uts.edu.au

ABSTRACT

‘The World of Hiroshige’ is an immersive virtual reality experience that allows the participants not only to view the artworks of the Japanese artist Utagawa Hiroshige in 3 dimensions, but also to experience and interact with the *Ukiyo* or “floating world” that inspired the works.

ACM Reference Format:

Alex Weight, Dylan Neil, and Daniel Flood. 2021. Virtual Reality Experience ‘The World of Hiroshige’: An Immersive Virtual Reality Experience of the World that Inspired the Works of the Japanese Artist Utagawa Hiroshige. In *SIGGRAPH Asia 2021 XR (SA ’21 XR)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3478514.3487611>

1 INTRODUCTION

Hiroshige’s famous Japanese woodblock prints known as *ukiyo-e*, depict life in the red-light district of Edo (modern-day Tokyo), and often feature geisha, samurai, prostitutes and villagers.

This project was developed by the Master of Animation and Visualisation students at the UTS Animal Logic Academy as part of their studies in emerging visualisation technologies. The project reaches beyond observational art virtual reality experiences in that it allows users to immerse themselves in the environment (see Figure 1), activities and customs of the time the artworks were created by using hand and body gestures to participate in archery, sailing, and interacting with animals and characters etc. Exploring the virtual environment also offers users specific views that align to some of the artist’s well-known works.

Characters, objects and the landscape were modelled in Maya, then surfaced to match patterns and designs from Hiroshige’s original works. The world and interactions were constructed in Unity. The project was originally developed as a clue-driven investigative experience, that could be adapted to any number of artworks. Further investigation examined additional style transfer methods that could be adopted for more painterly and abstract works.

This high-fidelity prototype and proof of concept can be expanded to other artists, art styles, works and worlds, with an ability to move between these. The immersive interaction can also be



Figure 1: Village scene from the interactive virtual reality project ‘The World of Hiroshige’. (UTS, 2020) (UTS, Landscape Construction, 2020)

expanded from a personal virtual reality to immersive spatial environments for broadscale interaction.

2 DESIGN AND IMPLEMENTATION

The environment was created using a hand-made depth map. Students painted a map of the island in Photoshop, using gradients to determine height and shape. This was then pushed through Unity’s Terrain Tools to create a 3D model (see Figure 2). Once that was completed and tested, to make sure everything was in the correct location, they then exported the 3D model and took it in to World Creator to paint in more detail around the edges of the lake and river. The environment was split into sections and given to multiple students to texture by hand. As the objective was to create a unique look that replicated Japanese prints, it was not possible to use auto generated surfacing tools. Instead, students hand-painted and surfaced each asset to match the style. The project was not intended to be a game, but rather an immersive experience. That being the case, all the interactive elements were designed to be calming and subtle. Eg. patting a horse, throwing rocks in the water etc.. Each of these elements highlighted through proximity detection when the user got closer, to indicate that they could interact (see Figure 3), this highlight also brought up a small instruction window showing them what to do. The sound design also played into this by playing a soft tone when the user got closer.

3 DISCUSSION

Although virtual reality art experiences have been produced before, this project goes beyond simply observing art in three dimensions,

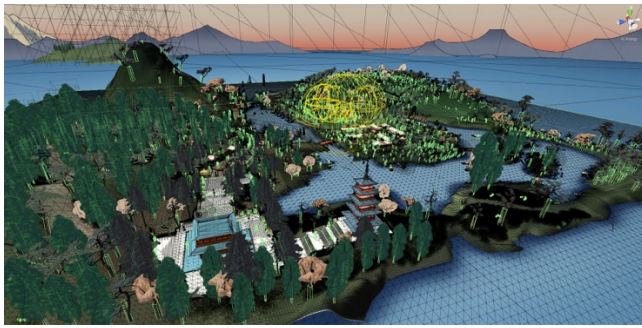


Figure 2: Landscape construction using Unity's Terrain Tools. (UTS, Landscape Construction, 2020)



Figure 3: Interactive characters become highlighted through proximity detection and sound design to suggest interaction through hand control and motion activation. (UTS, 2020)

but rather allows the user to experience the world that inspired the artist. Further research explored how other emerging visualisation techniques such as style transfer and eye tracking could be used for the interpretation of more abstract and painterly works which require a different type of interaction, as the vision for the development of the project extends beyond the one form of visual representation demonstrated here based on the works of Hiroshige.

Software and Hardware Implementation. Hardware: Any modern PC gaming computer with a Nvidia RTX2070 or greater. Oculus Quest 1 VR headset

Software: Windows 10, Oculus desktop app.

Technical Description. Uses a Oculus Quest 1 requiring USB tethering to a Desktop PC running Oculus desktop app in order to run the experience.

Space Requirement for Physical Demonstration: 1m x 1m

ACKNOWLEDGMENTS

This research project was completed by students from the UTS Animal Logic Academy's Master of Animation and Visualisation as part of their studies in emerging visualisation technologies.

Original concept by Mareena Iqbal, Sean Simon, Alexandra Moulard, Moses Benhardi The, Xuan Li, Wenhui Guo.

TECHNICAL DIRECTOR Sean Simon, PRODUCTION Eva Ly, Sean Simon, ART DIRECTION Victor Shergill, Vuong My Trinh,

PROGRAMING Sean Simon, Agape Paul Marundan, Ray Long, MODELING Gareth Sealy, Jason Capili, Rey La Torre, Ran Xie, Hernan Luzardo, Xiaoping Wang, Jessica Gu, Sol Cutrera, Wenhui Guo, Xinyu Deng, Jiacheng Wang, William Terry, Louis Green, SURFACING Xuan Li, Christopher Huang, Gareth Sealy, Hernan Luzardo, Xiaoping Wang, Jessica Gu, Jason Capili, Lauren Bartie, Moses Benhardi The, Semisi Lotulelei, Wenhui Guo, Louis Green, Xinyu Deng, William Terry, Jiacheng Wang, Max Mitchell, ANIMATION Alexandra Moulard, Mareena Iqbal, Semisi Lotulelei, Yuechen Du, Mathew Godfrey, Matt Jevris Fajardo, Xinyu Deng, RIGGING Moses Benhardi The, Lauren Bartie, Robbie Beresford, Xiaoping Wang, LAYOUT Mareena Iqbal, Yuechen Du, Mathew Godfrey, Matt Jevris Fajardo, Vuong My Trinh, Victor Shergill, Jessica Gu, William Terry, ART DEPARTMENT (UTS, 2020) Chelsea Hocroft, Max Mitchell, Jason Capili, Christopher Huang, SCORE Chelsea Marcelline, SOUND DESIGN & MIX Isaac Burton.

HEAD OF ACADEMY Ian Thomson, CREATIVE LEAD Alex Weight, VISUAL EFFECT LEADS Dylan Neill, Matt Estela, TECHNICAL LEAD Daniel Flood, PIPELINE TD Naomi Que, PRODUCTION TECHNOLOGY Andrew Lai, Seng Venly Ung, STUDIO COORDINATOR Sarah Giddy, COURSE DIRECTOR Professor Andrew Johnston.

Project URL: <https://animallogicacademy.uts.edu.au/project/world-of-hiroshige/>